

using empirical priors is difficult, but might still provide insights. When examining a generated material, in addition to performing a literature review, additional factors can be considered, including:

1. Symmetry; crystals are defined by their symmetry, and nature prefers symmetrical crystals unless there is a specific mechanism by which symmetry is broken. While lower symmetry would be expected as the number of elements in a system increases, in general generated crystals are expected to be symmetrical. P1 crystals are rare in nature, and when they are reported in databases this is often either because they have not been refined or even because of mis-identification [86].
2. The presence of defects. Defects could include structural distortions, with a crystal containing the “correct” atoms but in distorted geometry, or could include point defects, such as vacancies or interstitial atoms. The presence of defects is not necessarily bad; for example, many “off-stoichiometric” materials such as NiO_x are routinely synthesized which can contain large concentrations of vacancies. Sometimes vacancies might be required so that a structure might charge balance; a classic example might be the bixbyite crystal structure derived from a fluorite structure with an ordered array of structural vacancies. However, the presence of defects could also be a concern, especially in the case of unexplored chemical systems where a material might be erroneously calculated to be “on hull” due to incomplete knowledge of that chemical system.
3. Local atomic environments should be reasonable, meaning that the material contains reasonable bond lengths and co-ordination polyhedra, where other materials with those combinations of elements are known.
4. The material should charge balance if highly ionic, meaning that the sum of the formal valence of its constituent atoms is zero. If a material does not charge balance, and is ionic, it will likely have a very low defect formation energy if synthesizable. However, the importance of charge balance should be taken with caution, since the proportion of new materials that have been discovered that are nominally charge-balanced has decreased over time [87], with only around 40% newly-discovered materials being charge-balanced compared to over 80% of materials discovered a hundred years ago.

The definition of a material used in this work (Section 2) also allows for many potential failure modes including but not limited to: non 3D periodic materials such as 2D materials that contain a vacuum in one dimension, amorphous materials, or other random arrangements of “atoms-in-a-box”. Effort can be made to avoid these classes of materials by altering the training data of the generative model. Some efforts have been made to algorithmically categorize crystals [88], which could then be used to this effect, however these tools are not yet well-developed, and some spurious structures in the training set should be expected. Finally, the definition of a material used in this work also explicitly assumes an ordered material, whereas many real materials exist as alloys or solid solutions: in traditional DFT, only fully ordered materials can be calculated (meaning, a material that contains whole atoms, with exactly one atom on a given atomic site). This is also the constraint placed by MatterGen on its generated

crystals. However, real materials are often disordered, with fractional atomic occupancy (on *average*, a given atomic site might contain, say, 50% of one atom, and 50% of another atom). A disordered material has many “ordered approximations”—small unit cells with the correct overall composition—that can represent the parent disordered material. As such, MatterGen might generate several ordered approximations of the same parent disordered material. In these cases, novelty will not be assessed correctly, and properties such as energy above hull might be misleading.

Given these factors, we acknowledge that there are limitations in materials discovery efforts that still require methods advancements to overcome. We restrict our confidence that structures we generate are synthesizable to the level of theory and computation we use in this work, in addition to the finite reference we use for the convex hull. We have attempted human-assisted evaluation of predicted structures using empirical priors to gain additional knowledge for how our model performs at generating synthesizable materials.

D.2.1 Visualization

Manual analysis of crystal structures can be influenced by how they are represented visually, for example the specific bonds that are drawn. In this work, crystal structures are visualized using Crystal Toolkit [89] with the CrystalNN [90] bonding algorithm, since this is known to give good results in most cases. A uniform atomic radius was used since a wide variety of chemical bonding is expected to be present, and no one type of atomic radii (covalent, ionic, etc.) can be assumed. When valences are indicated, these are formal valences assigned using heuristic methods in `pymatgen` [79]. With the exception of Fig. 1, all visualizations are of a $2 \times 2 \times 2$ supercell to ensure at least one full repeat of a crystal and its periodic images are shown. All visualizations are of crystal structures as-calculated, meaning they are not necessarily in their conventional setting, and therefore axes are not labelled.

D.3 Generating stable and diverse materials

This section provides supplementary information to the results in Section 2.2.

D.3.1 RMSD, stability, uniqueness and novelty

To evaluate the performance of a generative model on the task of unconditional generation, we look at two keys metrics. First, we use the RMSD between the generated and the DFT-relaxed structure to measure local stability. Second, we use the fraction of S.U.N. structures to capture global stability and, to some extent, diversity. The RMSD metric is defined as

$$RMSD = \sqrt{\min_{\mathbf{P}} \frac{1}{N} \sum_n^N \left| \tilde{\mathbf{x}}_{\mathbf{P}(n)}^{gen} - \tilde{\mathbf{x}}_n^{DFT} \right|^2}, \quad (\text{D47})$$

where $\tilde{\mathbf{x}}_n$ indicates the Cartesian coordinates of atom n , and $\mathbf{P}$ is the element-aware permutation operator on the atoms of the generated structure. A lower RMSD indicates that generated structures are closer to their DFT-relaxed counterpart. This in turn reduces the computational time for the DFT relaxation, which is typically the most costly part of crystal structure generation. Novelty and uniqueness are computed in all model evaluations by comparing the atomic arrangement of every pair of structures that have the same reduced formula and space group via the `StructureMatcher` utility from the `pymatgen` Python package [79], with the following default parameters: `ltol=0.2`, `stol=0.3`, `angle_tol=5`. This definition of novelty is not able to determine whether an ordered structure might be an ordered approximation of a disordered structure, and so some structures might be falsely determined to be novel in this scenario.

The plots displayed in Fig. 2(b,c,e,f), and the structures showed in Fig. 2(1) refer to samples of 1024 generated structures which have also been relaxed via DFT.

D.3.2 Additional qualitative analysis of structures

Within the 1024 structures generated unconditionally for Fig. 2(a), a total of 43 unique, on-hull crystal structures were found: 11 binaries, 22 ternaries, and 10 quaternaries. These are summarized in Table D2. Of these, 3 had P1 symmetry, and 3 contained molecules or were molecular crystals. Prototype assignment limited to prototypes available in the `robocrystallographer` [91] tool, which will assign the “closest” matching prototype subject to tolerances. The ability for a composition to charge balance assessed by `pymatgen` and known common oxidation states for each element. As previously discussed, a generated crystal can still be reasonable even if it does not charge balance, and not all materials are ionic.

Four randomly-selected examples were highlighted in Fig. 2 in the main text. These were BaLa_2Ir , K_3AlCl_6 , NaNiO_6 , and $\text{NaSmTm}_2\text{Te}_4$. For BaLa_2Ir , it is well-known that La_2Ir forms an intermetallic with a Laves structure, and that Ba can often substitute for La since both can exist in a +2 oxidation state with a 6s2 outer shell. However, in this example, we see octahedrally co-ordinated Ir bonded to La, with Ba inserted as a single plane of atoms in a close-packed configuration, as it would exist in elemental Ba. It is unclear if this material could exist. The K_3AlCl_6 has low space group symmetry ($\text{P}\bar{1}$), but consists of Al in an ideal octahedral co-ordination, with K in a mixed bonding environment. This structure charge balances under the assumption of Al^{3+} , K^+ and Cl^- , and could be thought of as derived from a defected rocksalt. The NaNiO_6 structure exists in the training set and is well-known belonging to a family of periodate structures AMIO_6 (where A is an alkali metal and M is another metal atom). This material is therefore an example of a material incorrectly classified by our novelty filter; a material might be classified as novel prior to DFT relaxation, and not after relaxation. $\text{NaSmTm}_2\text{Te}_4$ is a rocksalt structure with Te in the anion site and a mix of Na, Sm and Tm in the cation sites.

D.4 Generating materials with target chemistry

This section provides supplementary information to the results in Section 2.3.

Formula	Symmetry	# Elements	Charge balances	Prototype	Classification
Ba ₄ Au	C2/m	2	-	-	Crystal with 1D chains
SbAu	P6 ₃ /mmc	2	Yes	Molybdenum Carbide MAX Phase	Bulk crystal
ReF ₆	Im $\bar{3}$ m	2	Yes	Tungsten	Molecular crystal
Tb ₂ Zn ₁₇	R $\bar{3}$ m	2	-	-	Bulk crystal
SeS	P1	2	Yes	red selenium	Crystal with 1D chains
V ₃ Cl ₈	C2/m	2	Yes	-	Layered/2D crystal
KCl ₈	P1	2	-	-	Hybrid
AlI ₇	P1	2	-	-	Crystal with 1D chains
VBr ₅	Cm	2	Yes	Silicon tetrafluoride	Molecular crystal
Li ₄ Hg	I4/m	2	-	-	Bulk crystal
DyIn ₃	Pm $\bar{3}$ m	2	-	Uranium Silicide	Bulk crystal
HfAlAu	P6 $\bar{2}$ m	3	-	-	Bulk crystal
Lu ₂ AgOs	P4/mmm	3	-	Heusler	Bulk crystal
GdScBi	P4/nmm	3	-	Matlockite	Bulk crystal
Sm(FeC) ₂	Fddd	3	Yes	-	Bulk crystal
Li(AlPd) ₂	P4/mbm	3	-	-	Bulk crystal
YbNiSn ₂	Cmcm	3	-	-	Bulk crystal
Nd(GaPt) ₂	P4/nmm	3	-	-	Bulk crystal
LiPrAs	P6m2	3	-	-	Bulk crystal
Eu(AgSe) ₂	P $\bar{3}$ m1	3	Yes	-	Bulk crystal
Tl ₄ IrO ₆	C2/m	3	Yes	-	Bulk crystal
CsTe ₂ Pd	C2/m	3	Yes	-	Bulk crystal
Al ₂ PdI ₈	C2/m	3	Yes	Indium	Molecular crystal
CaTbCd ₂	P4/mmm	3	-	Heusler	Bulk crystal
Hf ₂ ZnMo	P4/mmm	3	-	-	Bulk crystal
Sb ₅ PPb ₂	Amm2	3	Yes	-	Bulk crystal
La ₆ SbTe ₅	C2/m	3	Yes	Caswellsilverite	Bulk crystal
CeTeAs	Pnma	3	Yes	-	Bulk crystal
NaH ₃ Pd	Pm $\bar{3}$ m	3	Yes	(Cubic) Perovskite	Bulk crystal
Er ₂ ZnNi ₂	Immm	3	-	-	Bulk crystal
YBeSi	P6 ₃ /mmc	3	-	-	Bulk crystal
TePb ₅ Cl ₈	C2/m	3	Yes	-	Bulk crystal
FeCoH ₂	P4/mmm	3	-	Caswellsilverite	Bulk crystal
Sc ₄ GaCu ₂ Rh	R $\bar{3}$ m	4	-	Heusler	Bulk crystal
La ₈ Os ₃ PdBr ₄	R $\bar{3}$ m	4	-	Caswellsilverite	Bulk crystal
TbCe(Ho ₂ Te ₃) ₂	Cm	4	-	alpha Po	Bulk crystal
Ba ₈ Sb ₃ PBr ₄	R $\bar{3}$ m	4	Yes	alpha Po	Bulk crystal
Cs ₂ KZnF ₆	Fm $\bar{3}$ m	4	-	(Cubic) Perovskite	Bulk crystal
Zn ₂ Ni ₆ BH	Fm $\bar{3}$ m	4	-	(Cubic) Perovskite	Bulk crystal
Cs ₂ AuIBr ₆	I4/mmm	4	Yes	-	Bulk crystal
LiTb ₃ (DySe ₃) ₂	C2/m	4	-	alpha Po	Bulk crystal
Ba ₈ Pd ₃ RhAu	P2/m	4	-	beta Vanadium nitride	Bulk crystal
Te ₂ MoWSe ₂	Cm	4	Yes	Molybdenite	Layered/2D crystal

Table D2: Summary information on 43 crystal structures that were calculated as being thermodynamically stable from a batch of 1024 crystal structures from an unconditional generation task.

D.4.1 Additional experimental details

We explore the capability of MatterGen to find novel stable crystals across the 27 chemical systems listed in Table D3. We group the systems in terms of how many elements they contain (ternary, quaternary, and quinary), and in terms of how many structures near the the convex hull were present in the reference Alex-MP-ICSD dataset (‘well explored’, ‘partially explored’, ‘not explored’). The latter classes are defined as follows:

- ‘well explored’: systems with the highest numbers of structures near the convex hull. We removed all structures belonging to ‘well explored’ systems from the training data set to assess the capability of our model to recover existing stable structures without having seen them during training.

- ‘partially explored’: systems that lie between the 30th and the 90th percentile of the distribution of chemical systems based on the number of structures they have near the convex hull. This class was designed to assess the capability of our model to expand known convex hulls. Therefore, we did not remove the existing data belonging to such systems from the training set.
- ‘not explored’: systems with no data near the convex hull. This class was designed to test our model in chemical systems where no structures on the hull are present in the reference dataset.

Here, we define ‘near the convex hull’ structures as structures whose energy per atom is between 0.0 and 0.1 eV/atom above the convex hull. For all three groups, we randomly chose nine ternary, nine quaternary and nine quinary chemical systems (see Table D3). Moreover, we replaced those chemical systems that had an overlap of more than two elements with another system to promote chemical diversity. The replacement was chosen randomly as well.

For this task, we fine-tune our base model on two properties: chemical system and energy above hull. We encode the latent embedding for the energy above hull and the chemical system as detailed in Appendix B.2.1 and Appendix B.2.2, respectively. Both properties are available for all structures in the training set of the base model. Therefore, the training set is used in full for fine-tuning. At sampling time, we condition on both an energy above the convex hull of 0.0 eV/atom, and on the chemical system we want to sample.

To compare the performance of MatterGen against substitution and RSS, we employ an MLFF (MatterSim, see Appendix D.1.4) to relax the generated structures, and then perform *ab initio* relaxation and static calculations via DFT (see Appendix C.2 for details). In particular, we perform the following steps: (1) generate structures, (2) relax structures using the MLFF, (3) filter structures for uniqueness, (4) select the 100 structures with lowest predicted energy above hull according to the MLFF, (5) run DFT on these structures. We report metrics only with respect to those structures. To allow for a fair comparison between our generative model and non-generative approaches, we employ the MLFF relaxation on a greater number of samples for the latter. For RSS, we sample 600,000 structures per chemical system according to the protocol described in Appendix D.4.1. For substitution, we enumerate every possible structure according to the algorithm detailed in Appendix D.4.1, which yields between 15,000 and 70,000 structures per chemical system. For MatterGen, we generate 10,240 structures per chemical system.

Random structure search details

For every chemical system, we performed two rounds of RSS using the `airss` [49] package. In each round, we generated 300,000 structures by sampling 100,000 structures across three different ranges of number of atoms per unit cell. We used the following non-overlapping intervals: 3-9, 10-15, and 16-20 for ternary systems; 4-10, 11-15, and 16-20 for quaternary systems; 5-11, 12-16, and 17-20 for quinary systems. For the first round, we used `airss` to propose structures without structural relaxation using `MINSEP` = 0.7-3 (minimum separation between atoms in Å) and `SYMMOPS` = 2-4 (number of symmetry operations). All proposed structures were relaxed using an